

SEDAR 26: Caribbean Queen Snapper, Silk Snapper, and Redtail Parrotfish

Silk Snapper

Data Workshop

LIFE HISTORY WORKING GROUP

- It will be important to develop regional sampling programs to collect age and growth data for silk snapper, queen snapper, and redtail parrotfish to estimate growth parameters essential to length-based analyses. Estimates of age-growth parameters are currently limited for the three species in question, therefore, it is essential to continue to build upon the existing published research.
- Regional data collection programs should also be designed to evaluate morphological conversion factors for each species. There is a lack of consistency in the units of measure for length among the studies reviewed by the LHWG. An important area of research will be to develop length-length conversion factors for the three species.
- Length-at-full vulnerability is an important input for length-based analyses. Expansion and improvement of the TIP program will be crucial for continued collection of species-specific size information, which be used to estimate length-at full vulnerability.

PUERTO RICO CATCH STATISTICS WORKING GROUP

- Commercial Landings Expansion Factor - all recommendations are in progress. Port samplers are visiting different fishing centers, collecting data of landings by trip, species and effort
- The working group also recommended that the uncertainty in the annual reported landings be characterized by computing the variance of the expansion factors and confidence intervals about the calculated total landings.
- Increasing the dockside sampling of recreational fishing trips in Puerto Rico to reduce the uncertainty in the catch estimates and 2) 20 extending / initiate MRIP's efforts in the US Virgin Islands to quantify the magnitude of recreational catches. In addition, recreational effort.
- The recreational statistics Program recommends increasing the minimum number of trip interviews to 130 for shore fishing, 200 for private boats and 90 for charter boats.
- There is an immediate need to develop sampling efforts to better identify and quantify discards in the commercial fisheries.

USVI CATCH STATISTICS WORKING GROUP

- Initiate MRIP's efforts in the US Virgin Islands to quantify the magnitude of recreational catches.
- It is important to determine the efficacy of expansion factors used to estimate total catch. The information used to calculate expansion factors by year need to be verified.

- The collection of landings statistics in the U.S.V.I. should be species-specific because analysis of the current species-groupings is not informative for stock assessments. Species composition from TIP is not appropriate, given the current sampling methodology, for estimating species-specific landings using ratio estimators.
- It is important to encourage fishermen to submit all the monthly catch reports, to submit reports for months when they do not fish, and to complete all the fields in the reports, since critical information such as effort, gear, and location fished are often missing or incomplete.

FISHERY INDEPENDENT RESEARCH

- Continuation of ongoing, long term research may provide additional information for future assessments.

INDICES OF ABUNDANCE WORKING GROUP

- Well-designed, systematic research programs are essential to providing the data necessary for effective management. Much of the research reviewed lacked the necessary sample sizes and regular (ongoing) data collection needed to construct an adequate time series of catch and abundance indices
- A commitment to long-term research and data collection is essential for effective management. Short-term research and data collection are not the solution to the data problems identified in this assessment. Long-term research and monitoring are necessary in the Caribbean, as in any other managed fishery
- Emphasis should be placed on the improvement of the TIP sampling program, as catch rate standardization, catch composition and size-frequency analyses will continue to rely upon this information. Fishery-independent surveys and the collection of other biological data, however, are extremely important to develop alternative indices of abundance.
- Need to continue efforts to develop partnerships with local fishermen to conduct research and to collect needed data. Partnerships with the fishing community and other stakeholders are a cost effective way to collect components of the data necessary for the assessment process

Assessment Workshop

Research efforts are needed that focus on improved data collection efforts, particularly on better quantification of trip based catch and effort and recording of more detailed geographical data on catch area. Surveys should be considered that will allow validation of fisher reported catch, landings, and trip effort. Surveys are needed that allow characterization of multi- species trips to allow identification of trips that split fishing effort across different gears and species groups. These surveys should be coordinated with fisher groups to enhance support by the industry. The ability to characterizing trip specific CPUE data by life history stage is needed to aid in quantifying size or age specific abundance trends.

Length-frequency data and the corresponding age-growth relationships will likely serve as the primary mechanism to inform assessments in the US Caribbean in the near future. A direct focus on

increasing sampling intensity with a well-designed program (as being developed in the current Caribbean Data Improvement Plan) should be placed as a top priority. This is particularly important in the Virgin Islands where sample size issues alone preclude drawing conclusions from the current TIP data.

Reliable estimates of the von Bertalanffy growth parameters, required inputs for length-based mortality estimators, should be another research priority. For silk snapper, a recent age-growth relationships derived from data from the region would also allow stronger conclusions to be drawn from the TIP data. A well designed age and growth study should be instated to meet this research objective.

Review Workshop

Major priorities

1. There is large degree of uncertainty in the assessment due to the data poor nature of this fishery. In the short to medium terms, the key data set is likely to remain size frequency distributions. The ability to utilize length-frequency data is contingent upon having reliable estimates of life history parameters (von Bertalanffy parameters in particular), therefore the highest priority for future research are:
 - a. Studies on basic life history (e.g. age-growth relationships and estimating natural mortality) are essential in the US Caribbean and will greatly enhance the utility of the existing length-frequency data. This information should provide the greatest benefit to providing management advice in the short term. This should be placed as a top priority for key species.
 - b. At present, the TIP size frequency data provides the only source of information on stock status and benchmarks and it is therefore essential that this program be at least continued. However, expansion (for example, to USVI) and improvement of the TIP program will be recommended for continued collection of species-specific size information.
 - c. Focus should be on developing more complete and accurate data sets into the future, particularly on trip based catch and effort and recording of more geographical data on catch location.
 - d. The recreational catch and effort is an important data set and should be continued. Expanding this system to the USVI may also be useful. Furthermore, this source of mortality should be included in the analyses.
 - e. Emphasis should be placed on extension, as compliance and unreporting is likely to increase when more data is required of fishers. Given the present low rate of reporting in Puerto Rico, this would be of great concern.
 - f. Validation of fisher reported catch, landings and trip effort should be undertaken.
 - g. The collection of landings statistics in the USVI should be species-specific because analysis of the current species-groupings is not informative for stock assessments, unless future assessments and management action focus on logical clusters of species.

- h. Characterization of multi- species trips to allow identification of trips that split fishing effort across different gears and species groups. This work should be coordinated with fisher groups to enhance buy-in by the industry.
 - i. It is important to encourage fishermen to submit all the monthly catch reports (USVI), to submit reports for months when they do not fish, and to complete all the fields in the reports, since critical information such as effort, gear, and location fished are often missing or incomplete.
- 2. All sources of mortality should be considered in the analyses especially for the recreational fishery catch in Puerto Rico for Silk and Queen Snapper.
- 3. Given the importance of the SEINE method and that extensions of this method are likely to be used into the near future, the following additional modification are required:
 - a. When the full likelihood surface for the SEINE analyses were shown in session, it was clear that unnecessary combinations are sampled and that the surface is reasonably flat near the optimal likelihood, which means more sampling needs to be undertaken within this range.
 - b. The SEINE method should be extended to apply a Bayesian hierarchical model that draws on species with more information (Punt *et al.*, 2011, although this method is not Bayesian). This method would integrate across all the different forms of uncertainty and also allow more data rich species' information to be drawn from for the data poor species.
 - c. The SEINE method should be extended to include the estimate of M for those species where this information is available. . This directly acknowledged the correlation between growth, maximum length and natural mortality.
 - d. The SEINE method should be tested in a simulation study using a simulated population with known parameters, recruitment, and size frequency and including variability in key parameters. Furthermore, these results should then be converted to a guideline on how to apply this information in a data poor situation.
 - e. Some preliminary analyses were undertaken during the Review that should be further investigated.

Medium priority

- 1. For all landings series, a more appropriate method would be to present median estimates of landings with confidence intervals for all regions. All sources of uncertainty should be included in this analysis.
- 2. The CPUE standardisation methods needs much more extensive investigation, including:
 - a. The feasibility of including additional factors or variables either as offsets or ratios of catch to relevant species total catch should be undertaken in the future. An overall Redtailed Parrotfish index from the catch rate standardisation is developed in the future.
 - b. Developing an overall Redtailed Parrotfish index from the catch rate standardisation be developed in the future

3. Given the uncertainty in the data, any future FIS should be designed in such a way as to be aligned with the earlier surveys. This would be extremely useful for comparison.

Lower priority

1. There is some question whether changing the commercial catch expansion method during the series produces biases. Therefore, the effect of the two different methods over the time series to develop the expansion factors should be tested.
2. There is a need to develop sampling efforts to better identify and quantify discards in the commercial fisheries.

Queen Snapper

Data Workshop

LIFE HISTORY WORKING GROUP

- It will be important to develop regional sampling programs to collect age and growth data for silk snapper, queen snapper, and redbtail parrotfish to estimate growth parameters essential to length-based analyses. Estimates of age-growth parameters are currently limited for the three species in question, therefore, it is essential to continue to build upon the existing published research.
- Regional data collection programs should also be designed to evaluate morphological conversion factors for each species. There is a lack of consistency in the units of measure for length among the studies reviewed by the LHWG. An important area of research will be to develop length-length conversion factors for the three species.
- Length-at-full vulnerability is an important input for length-based analyses. Expansion and improvement of the TIP program will be crucial for continued collection of species-specific size information, which be used to estimate length-at full vulnerability.

PUERTO RICO CATCH STATISTICS WORKING GROUP

- Commercial Landings Expansion Factor - all recommendations are in progress. Port samplers are visiting different fishing centers, collecting data of landings by trip, species and effort
- The working group also recommended that the uncertainty in the annual reported landings be characterized by computing the variance of the expansion factors and confidence intervals about the calculated total landings.
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USVI CATCH STATISTICS WORKING GROUP

- Initiate MRIP's efforts in the US Virgin Islands to quantify the magnitude of recreational catches.
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effort across different gears and species groups. These surveys should be coordinated with fisher groups to enhance buy in by the industry.

The ability to utilize length-frequency data is contingent upon having reliable estimates of life history parameters (von Bertalanffy parameters in particular). Studies on basic life history (e.g. age-growth relationships and estimating natural mortality) in the US Caribbean will greatly enhance the utility of the existing length-frequency data and should provide the greatest benefit to providing management advice in the short term. This should be placed as a top priority for key species.

Review Workshop

Major priorities

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 - e. Emphasis should be placed on extension, as compliance and unreporting is likely to increase when more data is required of fishers. Given the present low rate of reporting in Puerto Rico, this would be of great concern.
 - f. Validation of fisher reported catch, landings and trip effort should be undertaken.
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Medium priority

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Redtailed Parrotfish

Data Workshop

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